

Education

PhD in Mathematics <i>Advisor:</i> Michael O'Neil <i>Thesis:</i> Fast transform methods for Gaussian random fields	<i>Courant Institute, New York University</i>	2020-2025
BS in Computational and Applied Mathematics <i>Advisor:</i> Mihai Anitescu <i>Thesis:</i> Nonstationary Gaussian process approximations of piecewise analytic computer codes	<i>University of Chicago</i>	2015-2019

Research Positions

Peter O'Donnell Jr. Postdoctoral Fellow <i>Advisors:</i> Joe Kileel, Per-Gunnar Martinsson	<i>Oden Institute, UT Austin</i>	2025-present
CSGF Practicum <i>Advisors:</i> Yang Liu, Xiaoye Sherry Li	<i>Lawrence Berkeley National Laboratory</i>	2023
Mathematics and Statistics Researcher <i>Advisors:</i> Mihai Anitescu, Michael Stein	<i>Argonne National Laboratory</i>	2019-2020

Publications

- Persson, David, **Paul G. Beckman**, Tyler Chen, Diana Halikias, Christopher Musco. "A Recursive Butterfly Factorization with Optimality Guarantees." arXiv preprint arXiv:2607.29361 (2026).
- Beckman, Paul G.**, Samuel F. Potter, Michael O'Neil. "A Butterfly-Accelerated Manifold Harmonic Transform." arXiv preprint arXiv:2605.21828 (2026).
- Beckman, Paul G.**, Michael O'Neil. "A Nonuniform Fast Hankel Transform." *SIAM Journal on Scientific Computing*, no. 2 (2026): A784-A803.
- Geoga, Christopher J., **Beckman, Paul G.**. "Nonparametric spectral density estimation from irregularly sampled data." arXiv preprint arXiv: 2503.00492 (2025).
- Beckman, Paul G.**, Christopher J. Geoga. "Fast Adaptive Fourier Integration for Spectral Densities of Gaussian Processes." *Statistics and Computing* 34, no. 6 (2024): 217.
- Beckman, Paul G.**, Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable Computations for Nonstationary Gaussian Processes." *Statistics and Computing* 33, no. 4 (2023): 84.
- Williams-Young, David B., **Paul G. Beckman**, and Chao Yang. "A Shift Selection Strategy for Parallel Shift-Invert Spectrum Slicing in Symmetric Self-Consistent Eigenvalue Computation." *ACM Transactions on Mathematical Software (TOMS)* 46, no. 4 (2020): 1-31.

Awards

Moses A. Greenfield Research Prize	<i>Courant Institute</i>	2024
Computational Science Graduate Fellowship	<i>Department of Energy</i>	2020

Presentations

CodEx Seminar "Fast transforms for Gaussian random fields"	<i>CodEx Seminar</i>	2026
UT Austin "Fast transforms for Gaussian random fields"	<i>Oden Institute Seminar</i>	2026
UT Austin "Implicit tensor moment estimation of nonparametric mixture models with sparse dependence"	<i>Data & Algebra Seminar</i>	2026

SIAM	<i>Uncertainty Quantification</i>	2026
"Fast transforms for spectral modeling of Gaussian processes"		
SIAM	<i>Computational Science and Engineering (Minisymposium co-organizer)</i>	2025
"Fast adaptive Fourier integration of spectral densities"		
Princeton University	<i>PACM IDeAS Seminar</i>	2025
"A nonuniform fast Hankel transform"		
SIAM	<i>Uncertainty Quantification</i>	2024
Talk: "Fast adaptive Fourier integration of spectral densities"		
Poster: "Butterfly-accelerated Gaussian random fields on manifolds"		
ICIAM	<i>International Congress on Industrial and Applied Mathematics</i>	2023
"Boundary integral methods for computing covariances in inverse source problems"		
SIAM	<i>Mathematics of Data Science (Minisymposium co-organizer)</i>	2022
"Fast algorithms for elliptic PDEs with Gaussian boundary noise"		

Teaching

Differential Calculus	<i>UT Austin M408K (TPEI)</i>	Co-instructor	Fall 2026
Mathematic Statistics	<i>NYU MATH-UA.2340</i>	Teaching Assistant	Spring 2024
Statistics	<i>NYU MATH-GA.2962</i>	Teaching Assistant	Fall 2021
Computational Statistics	<i>NYU MATH-GA.2080</i>	Teaching Assistant	Spring 2021

Software

Primary developer	FastHankelTransform.jl, SpectralKernels.jl
Contributor	FastGaussQuadrature.jl, chunkIE

Outreach and Service

Peer reviewer

Journals include *IEEE Signal Processing Letters*, *Statistics and Computing*, *Journal of Machine Learning Research*.

Volunteer Instructor	<i>Texas Prison Education Initiative (TPEI)</i>	2026-present
Co-instructor for credit-bearing college math courses for incarcerated youth and adults		
Volunteer Instructor	<i>Petey Greene Program</i>	2020-2025
Co-instructor for elementary through college level math for currently and formerly incarcerated adults		
Reading Group Founder	<i>Courant Institute</i>	2020-2024
Co-founded a department group reading articles on progress towards more accessible math higher education		
Department Climate Survey Designer	<i>Courant Institute</i>	2020-2022
Co-designed, administered, and reviewed a survey on Courant Institute PhD student experience		